



LUWA

Atmospheric intelligence for environmental decisions

luwaapp.com

<https://luwaapp.com/about.html>

Lucht-en
Waarneming
Analyse

On-demand chemistry transport modeling

- Emission estimation
- Sensitivity analysis to emission changes
- Deposition analysis
- Carbon footprint and emission factor estimation

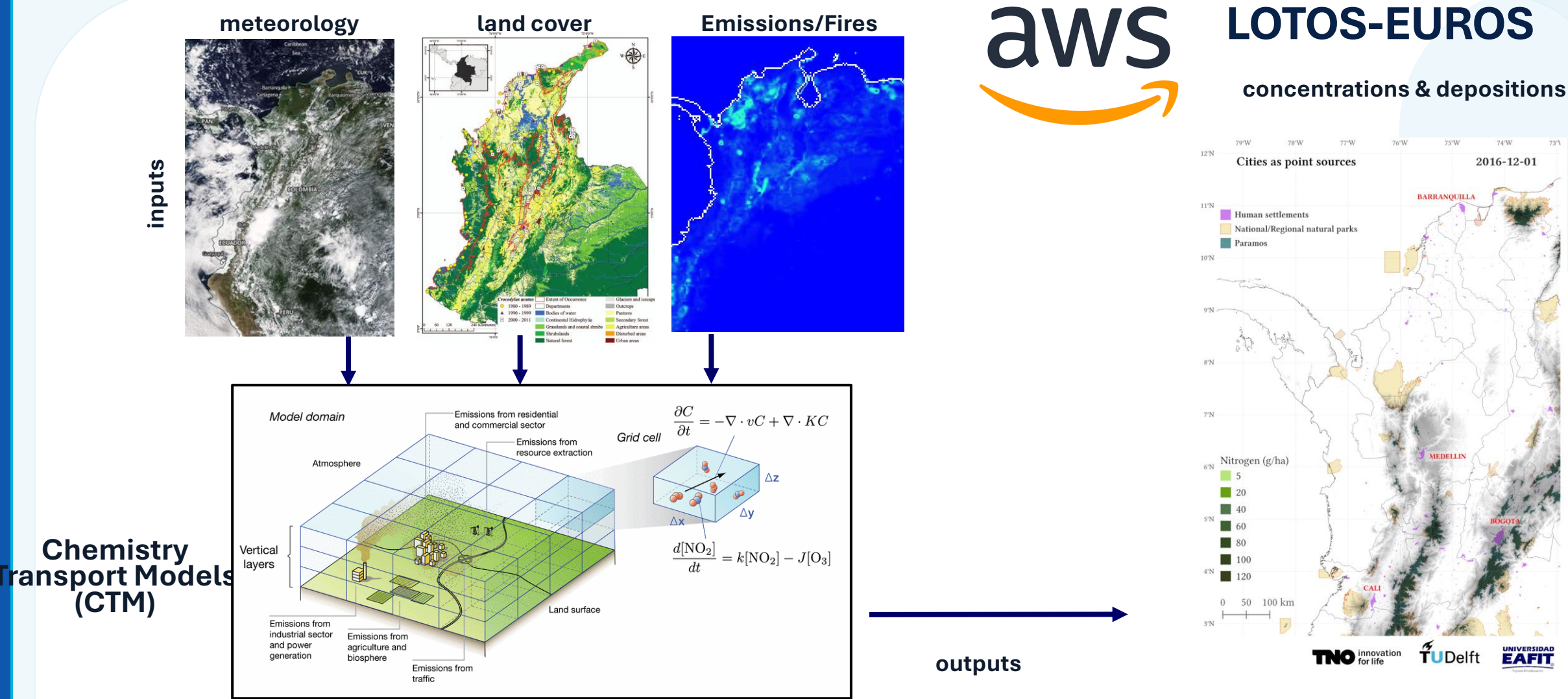
01

Deployment of sensors and IoT systems

Satellite data for atmospheric chemistry

On-demand chemistry transport modeling

- Emission estimation
- Sensitivity analysis to emission changes
- Deposition analysis
- Carbon footprint and emission factor estimation



Yarce Botero A., Montoya, O. L. Q., Lopez-Restrepo, S., Pinel, N., Hinestroza, J. E., Niño-Ruiz, E. D., ... & Lorduy, S. (2021). Medellin Air Quality Initiative (MAUI). In Environmental Sustainability-Preparing for Tomorrow. IntechOpen.

Yarce Botero, A., van Weele, M., Segers, A., Siebesma, P., & Eskes, H. (2024). Investigating the impact of coupling HARMONIE-WINS50 (cy43) meteorology to LOTOS-EUROS (v2. 2.002) on a simulation of NO₂ concentrations over the Netherlands. Geoscientific Model Development, 17(9), 3765-3781.



Laboratorio
Unificado de la
Atmósfera

Simulation request and data-delivery workflow

API interaction (user)

Asynchronous simulation workflow (system)

01 Authentication (API key)

```
ayarceb@AXARCEB: ~$ curl -X GET https://api.luwaapp.com/
HTTP/1.1 200 OK
Server: nginx/1.28.3 (Ubuntu)
Date: Tue, 15 Sep 2026 20:11:12 GMT
Content-Type: text/plain; charset=utf-8
Content-Length: 719
Connection: keep-alive

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Status: online
Environment: pilot
Base URL: https://api.luwaapp.com
Message: You are connected to the LUWA API.
Timestamp: 2026-09-15 20:11:12 UTC
```

Authenticate with your API key to connect to the LUWA API.

02 API endpoints

Method	Endpoint	Description
GET	/	Service root.
GET	/status	Simple status.
GET	/api/v1	API root v1.
GET	/api/v1/welcome	Welcome message.
GET	/api/v1/health	Operational health.
GET	/api/v1/service-info	Pilot menu service info.
GET	/api/v1/catalog	Catalog index.
GET	/api/v1/catalog/domains	Catalog index.
GET	/api/v1/catalog/domains/{domain_id}	Spatial domains.
GET	/api/v1/catalog/resolutions	Resolution details.
GET	/api/v1/catalog/variables	Variables.
GET	/api/v1/catalog/output-formats	Output formats.
GET	/api/v1/data/point	Point query.
GET	/api/v1/data/area	Rectangular query.
POST	/api/v1/simulations/submissions	Simulation request.

Use the API endpoints to explore available options and submit your simulation request.

03 Simulation request file

```
GNU nano 8.7.1 luwa-sim-20260910T181027Z-c5083b58.txt
[LUMA_SIMULATION_REQUEST]
simulation_id=luwa-sim-20260910T181027Z-c5083b58
submitted_at=2026-09-10T18:10:27Z
correlation_id=91bc553e-4d24-bb57-a299-d182f2027d6e

[Domain]
lat_min=20.3301
lat_max=20.861
lon_min=100.762
lon_max=99.9975

[Resolution]
resolution_km=20

[Date]
date_ini=2024-07-03T00:00:00Z
date_end=2024-07-26T00:00:00Z

[Variables]
variables=temperature_2m,wind_u_10m,wind_v_10m

[Email]
email=miguel_qs1999@gmail.com

[Workflow]
status=pending_manual_review
action=LUWA staff must verify this request and launch the
simulation. download_token=6f0a7256v6L6PRBdqnLuvLqySRKr9
```

Create and submit your simulation request file with the desired configuration (parameters, domain, dates and variables).

04 Email confirmation to the user

Hemos recibido tu solicitud de simulación

 **service@luwaapp.com**
para mi

Hola,

Hemos recibido correctamente tu solicitud de simulación en LUWA.

Identificador: luwa-sim-20260910T181027Z-c5083b58
Estado: En revisión

Te notificaremos por correo electrónico cuando la simulación sea iniciada.

Gracias por utilizar LUWA
(Laboratorio Unificado de la Atmósfera).

— El equipo LUWA

You will receive an email confirming that your simulation request has been received and is under review.

05 LOTOS-EUROS simulation in AWS




Your simulation is processed in AWS using LOTOS-EUROS.

The request is reviewed and, once approved, the simulation is executed in AWS using LOTOS-EUROS.

06 Email notification and download results

Tu simulación de LUWA está lista

 **service@luwaapp.com**
para mi

Hola,

Tu simulación (luwa-sim-20260910T181027Z-c5083b58) terminó sin problemas.

Puedes descargar los resultados aquí:
https://api.luwaapp.com/api/v1/downloads/Er0D30091QT-tc07HAG1BeWbb0N1q_oU

Este enlace es personal, por favor no lo compartas.

Gracias por usar LUWA.

— El equipo LUWA




You will receive an email when the simulation is finished, with a link to download the results (NetCDF files in a .zip archive).

Deployment of sensors and IoT systems

18 Sensors in **production** for



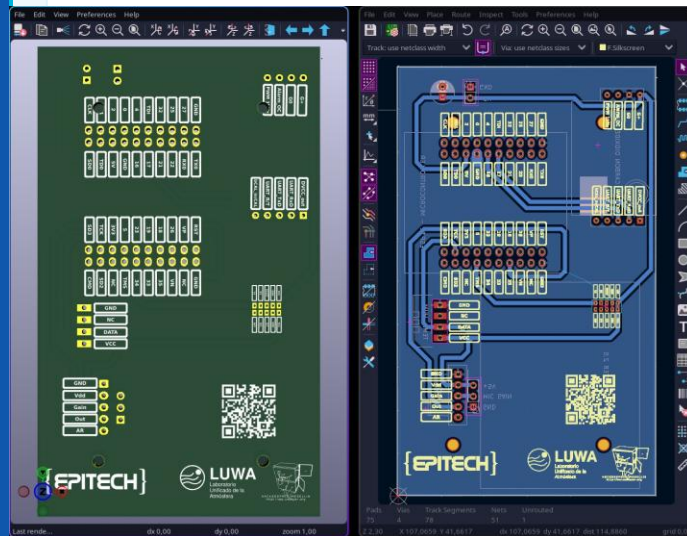
-Remote firmware update

-Plug and play philosophy

-Variables: H,T,(CO₂,NO₂,NH₃),PM₁₀,PM_{2.5},SP,

Next sensors: SO₂, Energy consumption: A,V

2 Sensor **operating** in Mexico Battery Lab



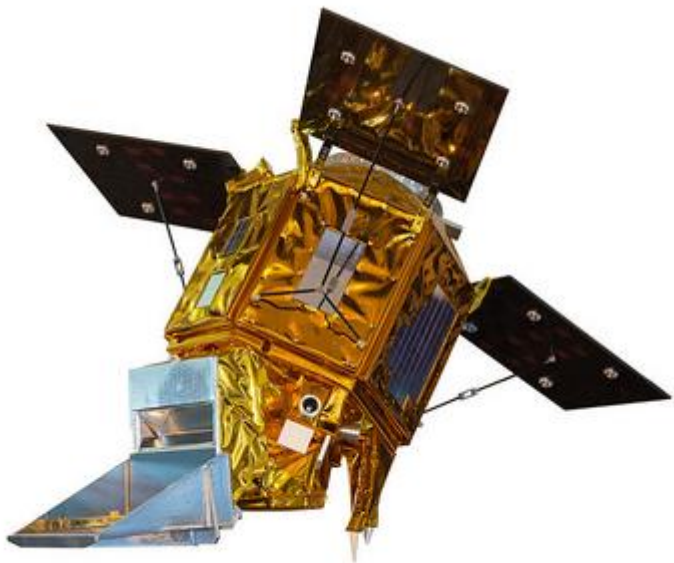
2 Sensor **operating** in Barcelona

8 Sensor **operating** in Medellín Colombia

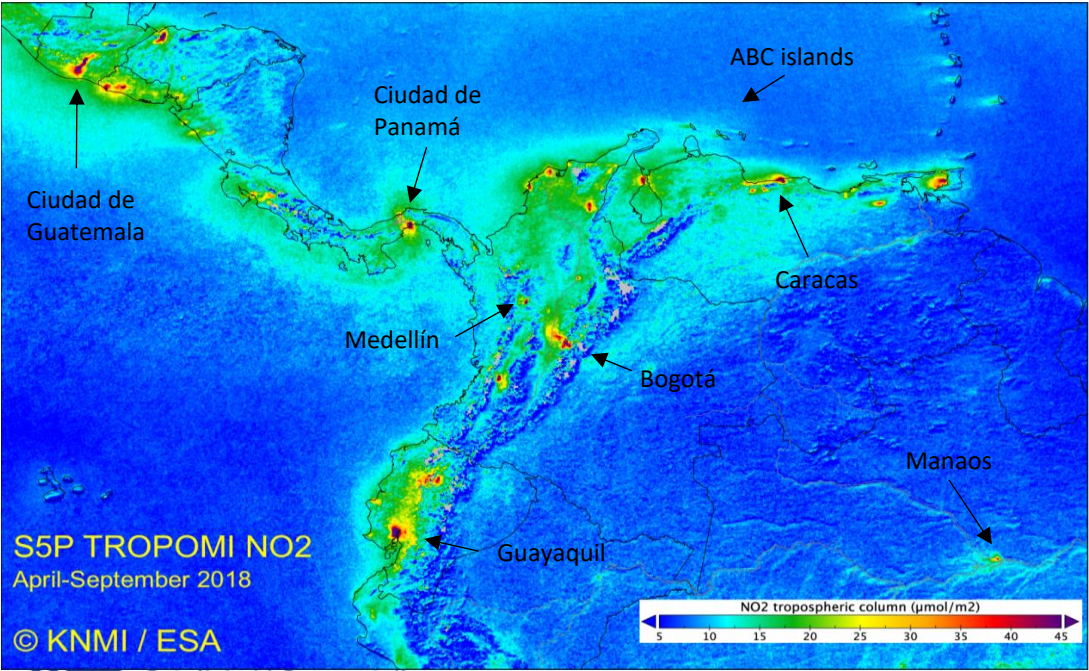
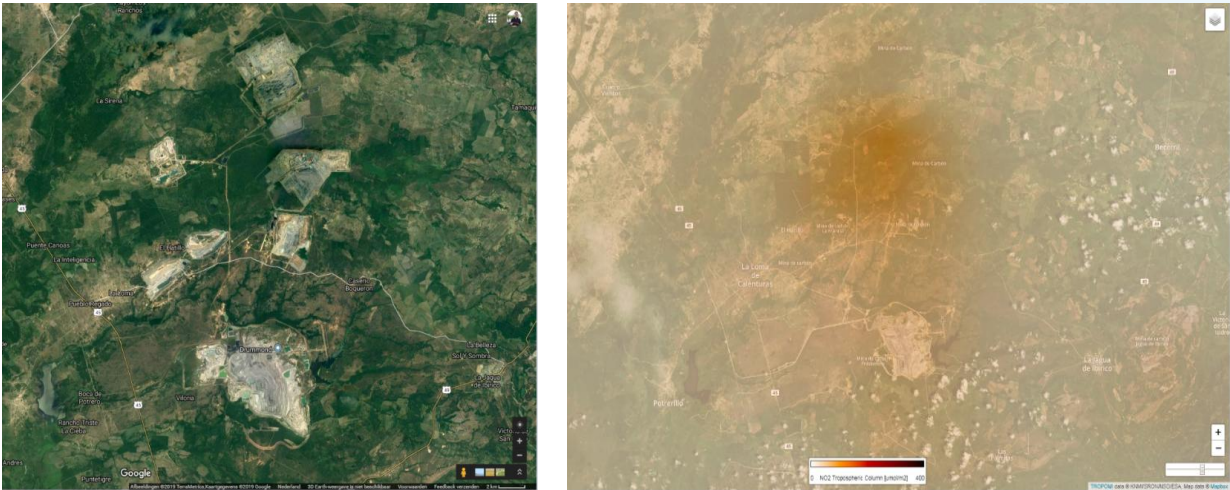


<https://luwaapp.com/sensing.html>

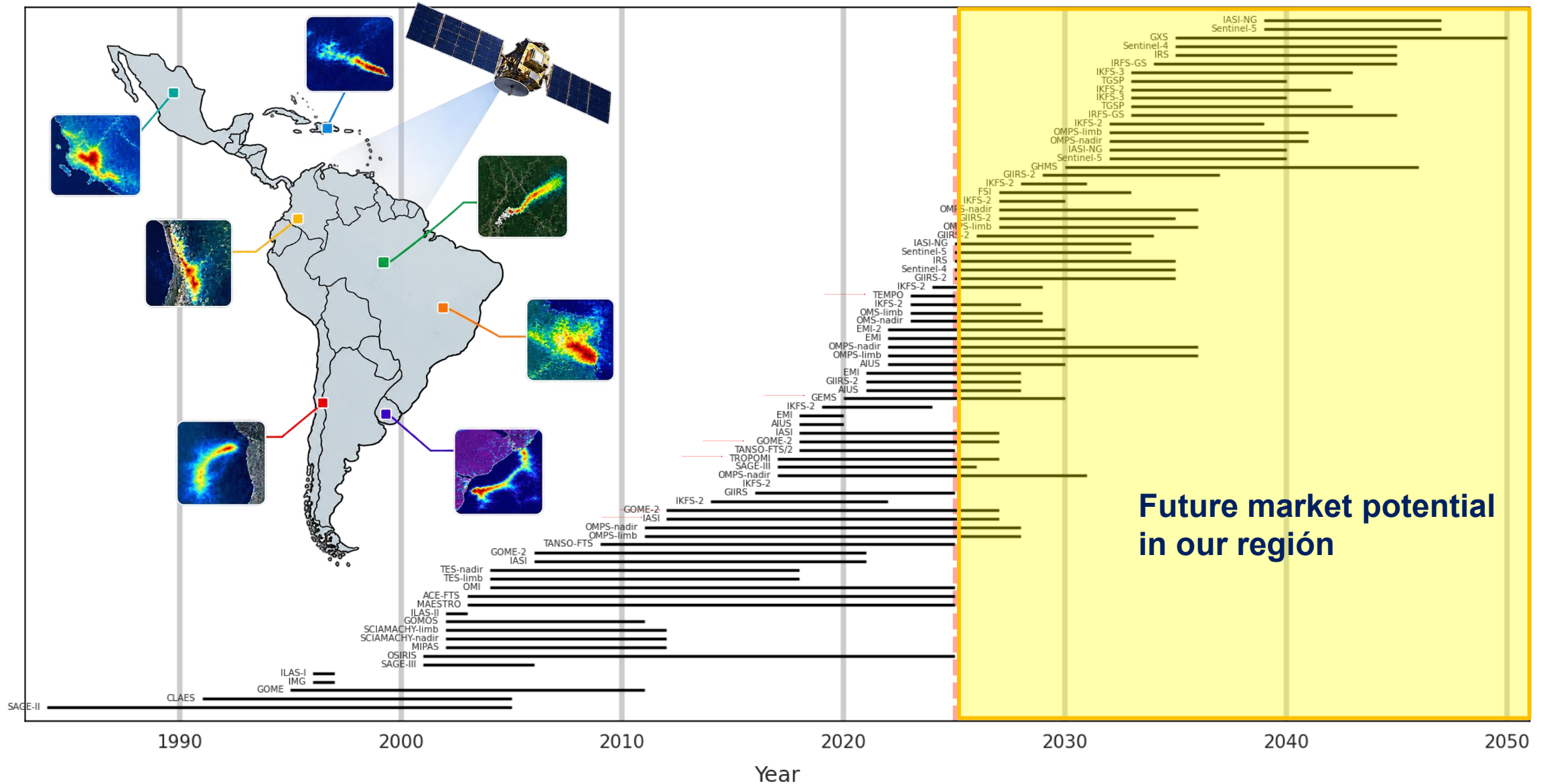
Satellite data for atmospheric chemistry



Open-pit mining activities, La Loma, April 2019



CSO CAMS Methodology



Plot based on data retrieved from <https://space.oscar.wmo.int/>



Open LUWA live

Cloud access · satellite observations · emission analysis · modeling outputs

<https://luwaapp.com/>

